

8 Day Report

Task: Azure ARM Deployment & Disk Configuration

Day: 8

Technology Used: Microsoft Azure (ARM Templates, VM, Managed Disks, Custom Script Extension)

Resource Group:

Objective of the Day

The objective of Day 8 was:

1. Deploy a Windows Virtual Machine using Azure ARM Template
 2. Configure networking components (VNet, Subnet, NSG, Public IP, NIC)
 3. Install IIS using Custom Script Extension
 4. Deploy a custom HTML page
 5. Create and attach a new managed data disk to the existing VM
 6. Verify disk configuration inside the virtual machine
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Task 1 – Deploy Windows VM using ARM Template

◆ Infrastructure Components Created

- Virtual Network (VNet)
- Subnet

- Network Security Group (NSG)
 - Public IP Address
 - Network Interface Card (NIC)
 - Windows 11 Virtual Machine
 - Custom Script Extension
-

◆ Network Configuration

Two inbound security rules were configured in NSG:

Port	Purpose
3389	RDP Access
80	HTTP Access (IIS)

This allowed:

- Remote Desktop access to the VM
 - Web access to the hosted HTML page
-

◆ Virtual Machine Configuration

- OS: Windows 11 Pro (Latest)
 - VM Size: Based on allowed SKU in subscription
 - OS Disk: Default Managed Disk
 - Authentication: Username & Password
-

◆ Custom Script Extension

A PowerShell script was uploaded to Azure Storage and executed using:

`Microsoft.Compute/virtualMachines/extensions`

Script Actions:

- Installed IIS (Web-Server Role)

Created custom HTML page in:

`C:\inetpub\wwwroot\index.html`

- - Restarted IIS service
 - Verified web hosting functionality
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◆ Output Verification

After deployment:

1. Public IP address was retrieved

Accessed in browser:

`http://<Public-IP>`

- 2.
 3. Successfully displayed custom IIS page
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Task 2 – Create and Attach a New Managed Data Disk

◆ Objective

To create a separate managed disk and attach it to the existing Windows VM.

◆ Disk Configuration

- Disk Type: Standard SSD (StandardSSD_LRS)
- Size: 64 GB
- Disk Name: `armDataDisk01`
- LUN: 0

◆ ARM Implementation

The template included:

`Microsoft.Compute/disks`

And attached to VM under:

`storageProfile → dataDisks`

◆ Inside VM Configuration

After deployment:

1. Connected to VM using RDP

Opened Disk Management:

`diskmgmt.msc`

- 2.
3. Initialized new disk
4. Created new simple volume
5. Formatted with NTFS

6. Disk appeared as D: drive
-

Challenges Faced

During implementation, several issues were encountered:

1 VM SKU Not Available

- Some VM sizes were restricted in the subscription
- Resolved by checking available SKUs

2 Authorization Error (RBAC)

Error:

```
AuthorizationFailed  
Microsoft.Resources/deployments/validate/action
```

Cause:

- Insufficient permissions (Contributor role required)

Resolution:

- Verified IAM permissions
- Adjusted deployment method

3 Deployment Conflicts

- Existing resources caused name conflicts
 - Resolved by modifying resource names
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Key Learnings

- Understanding Azure ARM template structure
 - Managing dependencies using `dependsOn`
 - Working with managed disks
 - Using Custom Script Extension
 - Handling RBAC permission errors
 - Troubleshooting SKU capacity issues
 - Managing LUN configuration for data disks
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Security Considerations

- NSG rules were restricted to required ports only
 - Standard SSD used for cost efficiency
 - Proper authentication configured
 - Deployment controlled via ARM templates
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Final Outcome

By the end of Day 8:

- ✓ Windows VM successfully deployed
 - ✓ IIS installed automatically
 - ✓ Custom HTML page hosted
 - ✓ Managed data disk created and attached
 - ✓ Disk initialized and formatted
 - ✓ Full infrastructure deployed using Infrastructure as Code
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Conclusion

Day 8 focused on practical implementation of Azure Infrastructure as Code using ARM Templates.

This session enhanced understanding of:

- Cloud Infrastructure Automation
- VM configuration
- Disk management
- Network security configuration
- Azure troubleshooting techniques

The objectives of the day were successfully achieved.

1 ARM Template – Create Windows Virtual Machine

 Save as: **vm-template.json**

```
{
  "$schema":
    "https://schema.management.azure.com/schemas/2019-04-01/deploymentTe
    mplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "vmName": { "type": "string", "defaultValue": "armVM01" },
    "adminUsername": { "type": "string", "defaultValue": "azureuser"
  },
  "adminPassword": { "type": "secureString" },
  "location": { "type": "string", "defaultValue":
    "[resourceGroup().location]" }
```

```
},  
"resources": [  
  {  
    "type": "Microsoft.Compute/virtualMachines",  
    "apiVersion": "2023-03-01",  
    "name": "[parameters('vmName')]",  
    "location": "[parameters('location')]",  
    "properties": {  
      "hardwareProfile": {  
        "vmSize": "Standard_B1s"  
      },  
      "osProfile": {  
        "computerName": "[parameters('vmName')]",  
        "adminUsername": "[parameters('adminUsername')]",  
        "adminPassword": "[parameters('adminPassword')]"  
      },  
      "storageProfile": {  
        "imageReference": {  
          "publisher": "MicrosoftWindowsServer",  
          "offer": "WindowsServer",  
          "sku": "2022-Datacenter",  
          "version": "latest"  
        },  
        "osDisk": {
```



```

        "createOption": "FromImage"
      }
    },
    "networkProfile": {
      "networkInterfaces": []
    }
  }
}
]
}

```

⚠ Note: This VM template assumes networking already exists.
If you need full networking included, tell me.

2 ARM Template – Create Storage Account

📁 Save as: **storage-template.json**

```

{
  "$schema":
    "https://schema.management.azure.com/schemas/2019-04-01/deploymentTe
    mplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "storageAccountName": {
      "type": "string"
    }
  }
}

```

```
    },  
    "location": {  
      "type": "string",  
      "defaultValue": "[resourceGroup().location]"  
    }  
  },  
  "resources": [  
    {  
      "type": "Microsoft.Storage/storageAccounts",  
      "apiVersion": "2023-01-01",  
      "name": "[parameters('storageAccountName')]",  
      "location": "[parameters('location')]",  
      "sku": {  
        "name": "Standard_LRS"  
      },  
      "kind": "StorageV2",  
      "properties": {}  
    }  
  ]  
}
```

- ◆ Storage type: General Purpose v2
 - ◆ Replication: Locally Redundant (LRS)
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3 ARM Template – Create Azure Container Registry (ACR)

 Save as: **acr-template.json**

```
{
  "$schema":
    "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "acrName": {
      "type": "string"
    },
    "location": {
      "type": "string",
      "defaultValue": "[resourceGroup().location]"
    }
  },
  "resources": [
    {
      "type": "Microsoft.ContainerRegistry/registries",
      "apiVersion": "2023-01-01-preview",
      "name": "[parameters('acrName')]",
      "location": "[parameters('location')]",
      "sku": {
```

```
      "name": "Basic"
    },
    "properties": {
      "adminUserEnabled": true
    }
  ]
}
```